

Modeling Brittle Materials Subjected to High-Velocity Impact

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This lecture presents recent computational work modeling brittle materials subjected to high-velocity impact. The focus will be on two brittle materials of significant interest: glass and a tungsten carbide cermet. In 2016 Holmquist and Johnson presented a constitutive model for glass for large strains, high strain rates and high pressures. This model is used to simulate the impact response of glass. Several example computations will be presented (and compared to experimental data) which will include: dwell, interface defeat, penetration, laser shock, edge-on impact, and multiple impacts onto the same glass target.

This lecture will also discuss the high-velocity impact response of a tungsten carbide (WC) cermet that is a component in the US Army Research Laboratory's (ARL) 14.5 mm BS41 projectile. The ARL BS41 is a complex projectile that includes a soft metal jacket, lead filler, inert powder and a high-strength WC core. Although the WC core is very brittle, it includes a small amount of cobalt binder (6 wt %) which produces a small amount of ductility in compression but none in tension. The addition of cobalt produces a cermet material (WC-.06Co) that exhibits a complex ballistic response. Advancements in 3D numerical algorithms and an accurate material model for the WC core produce computed results that are in good agreement with experimental data. Computed results will be presented that demonstrate the ability to reproduce several key experimental observations: the stripping of the steel jacket, lead filler and inert powder when impacting steel targets; rigid body penetration of the WC core into thin and thick steel targets for normal impact; severe fracture and fragmentation of the core when impacting obliquely; and core fracture due to a yaw angle at impact. Experimental data are also discussed and compared to the computed results.